

Cloud Computing

Question Bank — Model Answers

Units I – V | Covers 2, 4, 7 & 12 Mark Questions

UNIT I: Cloud Fundamentals

Q1. Define Cloud Computing. [2 Marks Marks]

Cloud computing is a model for enabling ubiquitous, convenient, on-demand network access to a shared pool of configurable computing resources (networks, servers, storage, applications and services) that can be rapidly provisioned and released with minimal management effort or service-provider interaction. In simple terms, it is the delivery of computing services over the Internet (the cloud) to offer faster innovation, flexible resources, and economies of scale.

Q2. List Characteristics of Cloud Computing. [2 Marks Marks]

According to NIST, cloud computing has five essential characteristics:

- On-demand self-service – Users can provision resources automatically without human interaction.
 - Broad network access – Resources are accessible over the network via standard mechanisms.
 - Resource pooling – Provider's resources serve multiple consumers (multi-tenant model).
 - Rapid elasticity – Resources can scale up or down quickly based on demand.
 - Measured service – Resource usage is monitored, controlled, and reported.
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Q3. State the NIST Model. [4 Marks Marks]

The National Institute of Standards and Technology (NIST) defines cloud computing through a formal model that encompasses three service models, four deployment models, and five essential characteristics.

Three Service Models:

- **IaaS (Infrastructure as a Service):** Provides virtualized computing infrastructure—virtual machines, storage, networks—over the Internet (e.g., AWS EC2, Azure VMs).
- **PaaS (Platform as a Service):** Provides a platform allowing customers to develop, run, and manage applications without managing the underlying infrastructure (e.g., Google App Engine, Heroku).
- **SaaS (Software as a Service):** Delivers software applications over the Internet on a subscription basis (e.g., Gmail, Microsoft Office 365, Salesforce).

Four Deployment Models:

- **Public Cloud:** Cloud services offered over the public internet and shared across organisations.
 - **Private Cloud:** Cloud infrastructure operated exclusively for a single organisation.
 - **Hybrid Cloud:** Combination of public and private clouds, bound together by technology.
 - **Community Cloud:** Shared infrastructure for specific communities with common concerns.
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Q4. Types of Deployment Models. [4 Marks Marks]

Cloud deployment models describe where the cloud infrastructure is located and who manages it. There are four primary deployment models:

1. Public Cloud

In a public cloud, infrastructure is owned and operated by a third-party provider and delivered over the Internet. Resources are shared among multiple tenants. Examples include Amazon Web Services (AWS), Microsoft Azure, and Google Cloud Platform. It offers high scalability and lower costs but less control over security.

2. Private Cloud

A private cloud is dedicated infrastructure operated solely for one organisation, either on-premises or by a third party. It provides greater control, security, and customization but requires higher capital expenditure. Examples: VMware vSphere, OpenStack-based private deployments.

3. Hybrid Cloud

A hybrid cloud combines public and private clouds, allowing data and applications to be shared between them. It provides greater flexibility and more deployment options. Organisations can keep sensitive workloads in a private cloud while using public cloud for burst capacity.

4. Community Cloud

A community cloud is shared by several organisations with common concerns (e.g., security, compliance, jurisdiction). It may be managed internally or by a third party. Example: Government agencies sharing a cloud for compliance reasons.

Q5. Define Cloud Computing Stack. [2 Marks Marks]

The cloud computing stack is a layered model that represents all the services and resources that make up cloud computing. It is structured as follows (bottom to top):

- Infrastructure Layer: Physical hardware — servers, storage, networking.
- Virtualization Layer: Hypervisors that abstract physical resources into virtual machines.
- IaaS Layer: Virtualized compute, storage, and network resources offered as a service.
- PaaS Layer: Development platforms, middleware, databases, and runtime environments.
- SaaS Layer: End-user applications delivered via the Internet.

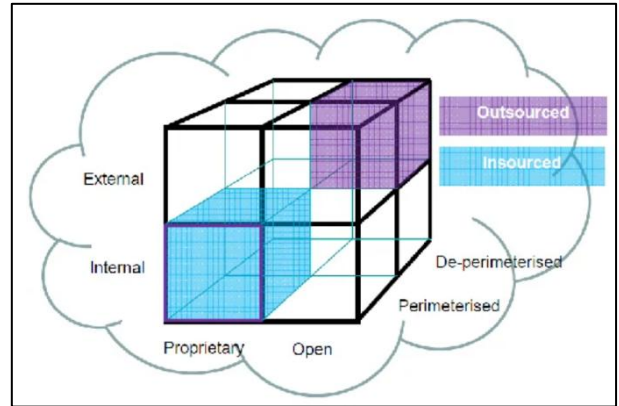
The stack concept helps understand how each layer builds on the one below it, with each layer consuming services from the layer beneath.

Q6. Explain the Cloud Cube Model. [7 Marks Marks]

The Cloud Cube Model was introduced by the Jericho Forum to help organisations make decisions about cloud computing in terms of information security. It represents four dimensions that describe different aspects of a cloud environment, forming a cube with eight possible configurations.

The Four Dimensions:

- **Internal / External:** Internal: Data and processing are within the organisation's own boundary. External: Data and processing reside outside the organisation's network perimeter, hosted by a third party.
- **Proprietary / Open:** Proprietary: Vendor-specific technology where the organisation is dependent on the vendor. Open: Uses open standards and interfaces, reducing vendor lock-in.
- **Perimeterised / De-perimeterised:** Perimeterised: Traditional security model using firewalls and a defined network boundary. De-perimeterised: Security travels with the data; boundaries are dissolved using technologies like VPNs or encryption.
- **Insourced / Outsourced:** Insourced: Cloud service is managed by the organisation itself. Outsourced: Managed and operated by a third-party service provider.



Significance:

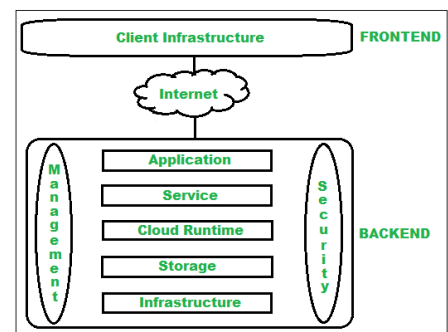
By plotting a cloud deployment along all four dimensions, organisations can assess factors like control, security, compliance, and flexibility. The model helps identify which combination of attributes best suits the organisation's risk appetite and business requirements. For example, an organisation handling sensitive financial data would prefer Internal + Proprietary + Perimeterised + Insourced for maximum control, while a startup might choose External + Open + De-perimeterised + Outsourced for agility and cost savings.

Q7. Draw and Explain Cloud Architecture. [7 Marks Marks]

Cloud architecture refers to the components and subcomponents that make up cloud computing. It describes the front-end and back-end platforms that communicate over a network (typically the Internet).

Key Components:

- **Front End:** The client side of the cloud computing system. It consists of the client's computer, network, and applications required to access the cloud (e.g., web browsers, mobile apps).
- **Back End:** The cloud itself, managed by the service provider. It comprises all the physical and virtual resources—servers, storage, virtual machines, databases, security mechanisms, and middleware.
- **Network / Internet:** Connects the front end and back end. The Internet is typically the primary medium, but private networks (WANs, VPNs) can also be used.
- **Cloud Platform:** The middleware and orchestration software that manages allocation of resources, monitors workloads, and maintains the service.
- **Hypervisor:** Virtualisation layer that allows multiple virtual machines to run on the same physical hardware.



- **Storage:** Scalable storage systems for persisting data (object storage, block storage, file storage).
- **Management Software:** Handles resource provisioning, metering, billing, and SLA enforcement.

How It Works:

A user (front end) sends a request via the Internet to the cloud provider's data centre (back end). The management layer authenticates the user, allocates resources, and routes the request to the appropriate compute node or storage system. The response travels back to the user. The entire process happens transparently and at high speed, giving the impression of virtually unlimited resources.

Q8. Differentiate Public, Private, and Hybrid Cloud [4 Marks Marks]

The following table summarises the key differences:

Feature	Public Cloud	Private Cloud	Hybrid Cloud
Ownership	Third-party provider	Single organisation	Both
Cost	Low (pay-per-use)	High (CapEx)	Moderate
Security	Lower	Higher	Varies
Scalability	Very High	Limited	High
Control	Low	Full	Partial
Example	AWS, Azure	VMware, OpenStack	AWS + on-prem

Q9. Explain Service Models (Overview) [4 Marks Marks]

Cloud service models define the type of service offered by the cloud provider to the consumer. There are three fundamental service models:

- **IaaS (Infrastructure as a Service):** Provides virtualised hardware resources over the Internet. Users manage the OS, middleware, and applications. The provider manages physical hardware, networking, and virtualisation. Example: Amazon EC2, Microsoft Azure Virtual Machines.
- **PaaS (Platform as a Service):** Provides a computing platform including OS, programming language runtime, database, and web server. Developers build and deploy applications without managing underlying infrastructure. Example: Google App Engine, Microsoft Azure App Service, Heroku.
- **SaaS (Software as a Service):** Delivers fully functional software applications over the Internet. The provider manages all layers—hardware, OS, middleware, and application. Users access via web browsers. Example: Gmail, Dropbox, Salesforce, Microsoft 365.

The key distinction is the level of management: IaaS gives most control to the user; SaaS gives the least.

Q10. Explain NIST Cloud Model with Diagram [7 Marks Marks]

The NIST SP 800-145 definition is the most widely accepted formal description of cloud computing. The NIST model defines cloud computing through three interconnected dimensions.

Five Essential Characteristics:

- **On-demand self-service:** Consumers can provision resources (storage, compute) automatically without requiring human interaction with each service provider.
- **Broad network access:** Resources are available over the network via standard mechanisms from heterogeneous client platforms (mobiles, laptops, workstations).
- **Resource pooling:** Provider's computing resources are pooled to serve multiple consumers using a multi-tenant model with physical and virtual resources dynamically assigned based on demand.
- **Rapid elasticity:** Resources can be elastically provisioned and released to scale rapidly outward and inward with demand.
- **Measured service:** Cloud systems automatically control and optimise resource use by leveraging a metering capability at some level of abstraction.

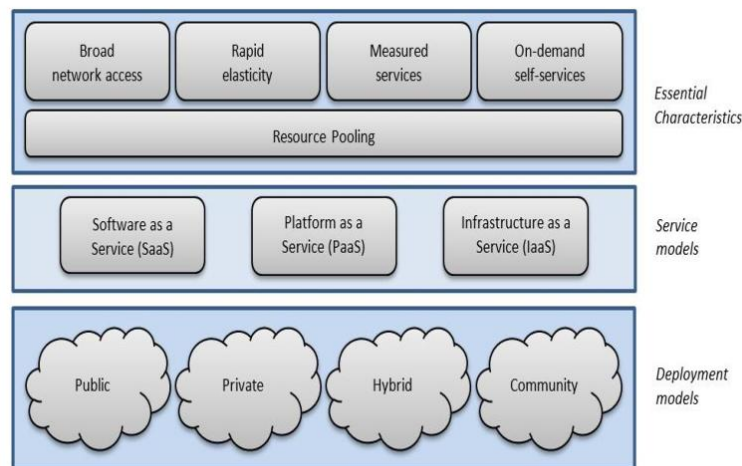
Three Service Models:

- **SaaS:** Software applications delivered over the Internet (e.g., Google Docs).
- **PaaS:** Application deployment and development platforms (e.g., AWS Elastic Beanstalk).
- **IaaS:** Virtualised computing infrastructure (e.g., Amazon EC2, Google Compute Engine).

Four Deployment Models:

Public Cloud, Private Cloud, Community Cloud, and Hybrid Cloud (described in detail in Q4 above).

Diagram Description (NIST Stack):



Q11. Describe Characteristics of Cloud Computing (Detailed) [7 Marks Marks]

The five NIST characteristics form the foundation of cloud computing. An additional set of characteristics are often discussed in the broader literature:

Core NIST Characteristics:

- **On-Demand Self-Service:** Enables automated provisioning—users access compute, storage, etc., via a portal or API without talking to a provider representative.
- **Broad Network Access:** All services are accessible from standard Internet connections using HTTP/HTTPS, ensuring device and platform independence.

- **Resource Pooling (Multi-Tenancy):** Physical resources are shared across multiple customers. Customers are isolated logically; each believes they have dedicated resources.
- **Rapid Elasticity:** Resources scale up (burst) during peak demand and scale down during off-peak periods—users pay only for what they use.
- **Measured Service:** Usage is tracked per user/application using metering. Billing is based on actual consumption (pay-per-use model).

Additional Characteristics:

- **Resilience and High Availability:** Cloud providers operate across multiple data centres with redundant hardware, ensuring near-zero downtime.
- **Cost Efficiency:** Converts capital expenditure (CapEx) to operational expenditure (OpEx)—no need to buy servers upfront.
- **Geographic Distribution:** Data and services can be replicated across multiple geographic regions for disaster recovery and latency reduction.

Q12. Explain Cloud Computing Architecture (Detailed) [7 Marks Marks]

A detailed cloud computing architecture involves several layers and components working together to deliver services seamlessly:

1. Client Layer (Front End):

Users interact with cloud services through clients such as web browsers, mobile apps, or thin clients. The client sends service requests over the Internet to the cloud infrastructure.

2. Network Layer:

The Internet or a WAN connects front-end clients to the back-end cloud infrastructure. Security mechanisms like firewalls, load balancers, and VPN gateways operate at this layer.

3. Service Layer:

This includes SaaS, PaaS, and IaaS offerings. APIs (RESTful or SOAP) act as interfaces so clients can programmatically request services.

4. Platform Layer:

Provides development frameworks, databases, messaging queues, and application servers. It sits between IaaS and SaaS.

5. Infrastructure Layer:

Physical servers, storage systems (SAN/NAS), and networking equipment reside here. Virtualisation software (hypervisors like VMware or Xen) abstracts this layer.

6. Data Storage Layer:

Includes object storage (Amazon S3), block storage (EBS), and distributed file systems (HDFS). Data is replicated across nodes for durability.

7. Management Layer (Orchestration):

Handles resource provisioning, auto-scaling, SLA management, billing, identity and access management (IAM), and monitoring. Tools include OpenStack, CloudStack, and vendor-specific management consoles.

Q13. Detailed Explanation of Cloud Computing Models (Deployment + Service)

[12 Marks Marks]

Cloud computing models are classified into two categories: Service Models (what is delivered) and Deployment Models (how it is deployed). Together they define how cloud computing is offered and consumed.

A. Service Models

1. Infrastructure as a Service (IaaS):

IaaS is the most basic category of cloud computing services. It provides virtualised computing infrastructure over the Internet. The provider manages physical servers, storage, and networking, while the customer manages the OS, middleware, and applications.

Key Features: VM provisioning, block and object storage, virtual networking, firewalls, auto-scaling.

Examples: Amazon EC2, Google Compute Engine, Microsoft Azure Virtual Machines, DigitalOcean Droplets.

Use Cases: Test/dev environments, website hosting, big data analytics, high-performance computing.

2. Platform as a Service (PaaS):

PaaS provides a cloud-based platform—including OS, runtime, database, middleware, and development tools—on which developers can build and deploy applications. The customer manages only the application and data.

Key Features: Application hosting, automated deployment, database management, scalability.

Examples: Google App Engine, Heroku, Microsoft Azure App Service, AWS Elastic Beanstalk.

Use Cases: Web application development, API development, IoT app backends.

3. Software as a Service (SaaS):

SaaS delivers software applications over the Internet on a subscription basis. The provider manages everything—infrastructure, platform, and application. Users access the software through a browser or thin client.

Key Features: No installation required, automatic updates, multi-tenant architecture, subscription billing.

Examples: Google Workspace, Microsoft 365, Salesforce, Zoom, Dropbox.

Use Cases: Email, CRM, ERP, collaboration, file storage.

B. Deployment Models

1. Public Cloud:

Infrastructure is owned by a third-party provider and shared among multiple organisations. Resources are provisioned over the Internet. Offers maximum scalability and cost efficiency. Security is managed by the provider. Examples: AWS, Azure, Google Cloud.

2. Private Cloud:

Infrastructure is dedicated to a single organisation, hosted either on-premises or by a third party. Offers greater security, control, and customization. Suited for organisations with strict regulatory requirements (healthcare, banking). Examples: OpenStack, VMware vCloud.

3. Hybrid Cloud:

Combines public and private cloud environments. Sensitive data stays in the private cloud; less critical workloads use the public cloud. Enables 'cloud bursting'—temporarily shifting workloads to the public cloud during peak demand. Examples: Azure Arc, AWS Outposts.

4. Community Cloud:

Shared by multiple organisations with common goals or requirements (e.g., government agencies, universities). Costs and management responsibilities are shared. Example: Government cloud initiatives shared across multiple departments.

C. Comparison of Service Models:

Aspect	IaaS	PaaS	SaaS
Managed by User	OS, App, Data	App, Data	Nothing
Managed by Provider	Hardware, Network	Hardware, OS, Runtime	Everything
Flexibility	Highest	Medium	Lowest
Cost Model	Pay per VM/hour	Pay per usage	Subscription
Example	EC2	App Engine	Gmail

UNIT II: Services & Virtualization

Q14. Define IaaS, PaaS, SaaS [2 Marks Marks]

- **IaaS:** Provides virtualised hardware resources (servers, storage, networking) over the Internet. Users manage the OS and above. Example: AWS EC2.
 - **PaaS:** Provides a platform for developing and deploying applications without managing infrastructure. Example: Google App Engine.
 - **SaaS:** Delivers complete software applications over the Internet on a subscription basis. Example: Gmail, Salesforce.
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Q15. Define Virtualization. [2 Marks Marks]

Virtualization is the process of creating a virtual version of a resource—such as a server, storage device, network, or operating system—using software. It abstracts physical hardware so that multiple virtual instances can run on a single physical machine. Virtualization is the foundational technology underlying cloud computing, enabling resource sharing, isolation, and efficient utilisation of hardware.

Q16. What is a Hypervisor? [2 Marks Marks]

A hypervisor (also called a Virtual Machine Monitor or VMM) is software that creates and manages virtual machines (VMs) on a physical host machine. It abstracts the physical hardware and allocates resources (CPU, memory, storage) to multiple VMs simultaneously. There are two types:

- **Type 1 (Bare Metal):** Runs directly on physical hardware (e.g., VMware ESXi, Microsoft Hyper-V, Xen).
 - **Type 2 (Hosted):** Runs on top of an existing OS (e.g., VMware Workstation, VirtualBox).
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Q17. Types of Virtualization [4 Marks Marks]

Virtualization can be applied across different layers of the IT stack:

- **Server Virtualization:** Multiple virtual servers run on a single physical server. Each VM has its own OS. Most common type in cloud computing. Example: VMware vSphere.
 - **Storage Virtualization:** Physical storage from multiple devices is pooled into a single virtual storage unit. Simplifies management and improves utilisation. Example: SAN, NAS virtualization.
 - **Network Virtualization:** Combines hardware and software network resources into a single, software-based administrative entity (virtual network). Example: VLAN, SDN.
 - **Desktop Virtualization (VDI):** Hosts desktop environments on a centralised server, accessed remotely by thin clients. Example: Citrix Virtual Apps, VMware Horizon.
 - **Application Virtualization:** Applications run in a virtual environment, isolated from the underlying OS. Example: Docker containers, Citrix XenApp.
 - **Memory Virtualization:** Aggregates RAM from multiple nodes into a shared memory pool accessible to all VMs.
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Q18. Difference Between SaaS and PaaS [4 Marks Marks]

Feature	SaaS	PaaS
Definition	Complete software delivered via Internet	Platform for building & deploying apps
User manages	Nothing	Application and data
Provider manages	Everything	Hardware, OS, runtime
Target users	End users	Developers
Customization	Very limited	High (code-level)
Example	Salesforce, Gmail	Google App Engine, Heroku
Pricing	Per user/month	Per usage/resource

Q19. Explain Hypervisors [7 Marks Marks]

A hypervisor is a critical component in virtualised and cloud environments. It serves as an abstraction layer between physical hardware and virtual machines.

Type 1 Hypervisors (Bare Metal):

Type 1 hypervisors run directly on the physical hardware without a host OS. They have direct access to hardware resources and provide better performance, stability, and security.

- **VMware ESXi:** Enterprise-grade hypervisor widely used in data centres.
- **Microsoft Hyper-V:** Integrated into Windows Server; supports live migration of VMs.
- **Xen:** Open-source hypervisor used by AWS in its early infrastructure.
- **KVM (Kernel-based Virtual Machine):** Linux kernel module that turns Linux into a Type 1 hypervisor.

Type 2 Hypervisors (Hosted):

Type 2 hypervisors run on top of an existing host OS. They are easier to install and use but have higher overhead because hardware calls pass through the host OS.

- **Oracle VirtualBox:** Free, open-source; supports Windows, Linux, macOS guests.
- **VMware Workstation:** Commercial product used for desktop virtualisation and testing.

Key Functions of a Hypervisor:

- VM creation, deletion, and lifecycle management
- CPU scheduling across multiple VMs
- Memory allocation and isolation between VMs
- I/O virtualisation (disk, network) via virtual device drivers
- Snapshot and cloning of VM state
- Live migration of running VMs to different physical hosts

Role in Cloud Computing:

Hypervisors enable the multi-tenant architecture essential to cloud computing. They ensure that each customer's VM is isolated from others, even when running on the same physical server. They also support resource pooling and rapid elasticity by allowing VMs to be created, resized, or migrated dynamically.

Q20. Explain Load Balancing [4 Marks Marks]

Load balancing is the process of distributing incoming network traffic or computing workloads across multiple servers or resources to ensure no single resource is overwhelmed. It is a critical feature in cloud environments to ensure high availability, reliability, and performance.

How It Works:

A load balancer sits in front of a pool of servers (backend instances). When a request arrives, the load balancer selects an appropriate server based on a configured algorithm and forwards the request. The client is unaware of which backend server handles the request.

Load Balancing Algorithms:

- **Round Robin:** Requests are distributed sequentially across servers in rotation.
- **Least Connections:** New requests go to the server with the fewest active connections.
- **IP Hash:** Client IP address determines which server handles the request (useful for session persistence).
- **Weighted Round Robin:** Servers with higher capacity receive proportionally more requests.

Types:

- **Layer 4 (Transport Level):** Routes based on IP and TCP/UDP port.
 - **Layer 7 (Application Level):** Routes based on content, URL, or HTTP headers. Example: AWS Elastic Load Balancer (ALB).
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Q21. Identity as a Service (IDaaS) [4 Marks Marks]

Identity as a Service (IDaaS) is a cloud-based identity and access management (IAM) solution delivered as a service. It provides authentication, authorisation, and user management capabilities to organisations without requiring them to build or maintain their own identity infrastructure.

Core Features:

- **Single Sign-On (SSO):** Users log in once and gain access to all connected applications without re-authenticating.
- **Multi-Factor Authentication (MFA):** Adds an extra verification layer (OTP, biometrics) beyond password.
- **User Provisioning/Deprovisioning:** Automates creation and deletion of user accounts when employees join or leave.
- **Directory Services:** Centralised directory of user identities (similar to Microsoft Active Directory, but in the cloud).
- **Federation:** Allows identity sharing across organisational boundaries (e.g., partner companies).

Examples: Microsoft Azure Active Directory (Entra ID), Okta, Auth0, Ping Identity.

Benefits:

Reduces IT overhead, improves security, enforces access policies, and supports compliance with regulations like GDPR and HIPAA.

Q22. Explain Virtualization and Its Types (Detailed) [7 Marks Marks]

Virtualization is the creation of a virtual (rather than actual) version of something, such as an OS, server, storage device, or network resource. It enables multiple virtual instances to share the same physical hardware, improving utilisation, flexibility, and cost efficiency.

How Virtualization Works:

The hypervisor (VMM) sits between the hardware and the VMs. It intercepts hardware calls from each VM, translates them, and executes them on the physical hardware. Each VM believes it has exclusive access to dedicated resources while the hypervisor multiplexes physical resources.

Types of Virtualization:

- **Full Virtualization:** Complete simulation of underlying hardware. Guest OS runs unmodified. High compatibility but slightly more overhead. Example: VMware vSphere.
- **Para-virtualization:** Guest OS is modified to communicate directly with the hypervisor via a special API (hypercalls). Better performance than full virtualisation. Example: Xen with modified Linux guests.
- **Hardware-assisted Virtualization:** CPU provides special instructions (Intel VT-x, AMD-V) that help the hypervisor manage VMs more efficiently without modifying the guest OS.
- **OS-level Virtualization (Containers):** Multiple isolated user-space instances run on a shared OS kernel. Lightweight and fast. Example: Docker, LXC.
- **Storage Virtualization:** Abstracts physical storage into a logical pool. Example: VMware vSAN, IBM SAN Volume Controller.
- **Network Virtualization:** Creates virtual networks independent of physical hardware. Enables software-defined networking (SDN). Example: VMware NSX, OpenStack Neutron.

Benefits of Virtualization:

- Improved hardware utilisation (consolidation)
 - Faster provisioning and deployment of new environments
 - Isolation and security between workloads
 - Disaster recovery through VM snapshots and live migration
 - Cost savings by running multiple workloads on fewer physical machines
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Q23. Role of Virtualization in Cloud Computing [4 Marks Marks]

Virtualization is the backbone of cloud computing. It enables the core characteristics of the cloud—on-demand provisioning, resource pooling, elasticity, and multi-tenancy—to be realised in practice.

- **Enables Multi-Tenancy:** Multiple customers can share the same physical hardware securely. Each customer's VM is isolated from others by the hypervisor.

- **Supports Resource Pooling:** Physical resources are abstracted into a shared pool. Cloud providers can dynamically allocate CPU, memory, and storage to VMs based on demand.
- **Enables Rapid Elasticity:** New VMs can be provisioned in seconds. Workloads can be scaled up or down by adding or removing VMs.
- **Facilitates Migration and High Availability:** VMs can be live-migrated from one physical host to another without downtime, enabling maintenance without service interruption.
- **Supports IaaS:** IaaS is essentially virtualised compute, storage, and network rented to customers. Without virtualisation, IaaS cannot exist.

Q24. Compare SaaS, PaaS, IaaS [7 Marks Marks]

All three service models deliver IT resources over the Internet but differ fundamentally in what is provided and what the customer manages.

Aspect	IaaS	PaaS	SaaS
Full Form	Infrastructure as a Service	Platform as a Service	Software as a Service
Who manages hardware	Provider	Provider	Provider
Who manages OS	User	Provider	Provider
Who manages app	User	User	Provider
Who manages data	User	User	Provider
Scalability	Manual / API	Auto + manual	Fully automatic
Target audience	Sys admins, IT teams	Developers	End users
Customisation	Very high	High	Low
Example	Amazon EC2	Google App Engine	Gmail, Salesforce
Pricing model	Pay per VM/hour	Pay per resource	Per user/month

The choice of model depends on the organisation's technical capabilities and how much control is needed. System administrators prefer IaaS for flexibility; developers prefer PaaS to focus on code; non-technical business users prefer SaaS for simplicity.

Q25. Detailed Explanation of Virtualization with Diagram [12 Marks Marks]

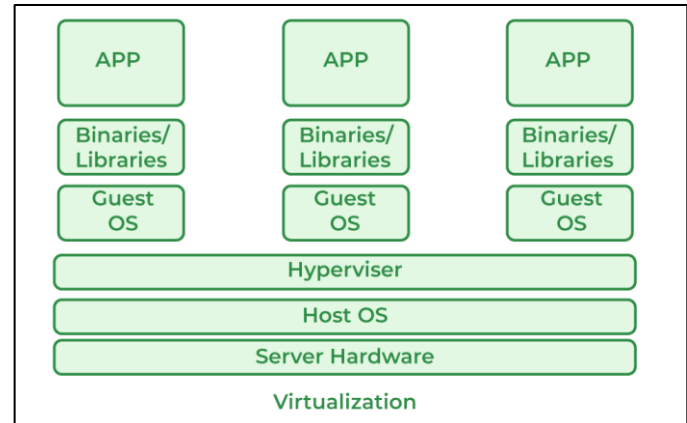
Virtualisation is the technology that enables cloud computing by abstracting physical hardware into logical, software-defined entities. A thorough understanding requires examining its architecture, types, implementation, and benefits.

1. Architecture:

The virtualization architecture consists of:

- **Physical Layer:** Actual hardware: CPUs, RAM, storage devices, NICs.

- **Hypervisor (VMM) Layer:** Software that intercepts and manages hardware access from VMs. Acts as the intermediary between physical hardware and VMs.
- **Virtual Machine Layer:** Each VM has virtual CPU, virtual RAM, virtual NICs, virtual disks. Each runs its own OS independently.
- **Guest OS and Application Layer:** Software running inside each VM, unaware it is virtualised.



2. How Virtualisation Works (Step-by-Step):

- Step 1: Hypervisor is installed on the physical hardware.
- Step 2: The hypervisor creates a pool of virtual hardware resources from the physical hardware.
- Step 3: A VM is created with a defined allocation of vCPUs, vRAM, and vDisk.
- Step 4: A guest OS is installed inside the VM.
- Step 5: When the guest OS executes privileged instructions, the hypervisor intercepts them and executes them safely on real hardware.
- Step 6: The hypervisor schedules VM execution on physical CPU cores.

3. Memory Virtualisation:

The hypervisor maps VM virtual addresses to machine physical addresses using Shadow Page Tables or hardware-assisted Extended Page Tables (EPT). This allows multiple VMs to share physical RAM safely.

4. I/O Virtualisation:

VMs use virtual network cards and disk controllers. The hypervisor translates virtual I/O requests to real hardware calls. Technologies like SR-IOV (Single Root I/O Virtualization) allow direct hardware access for performance-critical workloads.

5. Full vs Para vs Container Virtualisation:

- **Full Virtualisation:** Entire hardware emulated. No OS modification needed. Slightly lower performance due to binary translation.
- **Para-virtualisation:** Guest OS modified to use hypercalls (direct hypervisor API). Better performance. Used in Xen.
- **Containers:** OS-level virtualisation sharing the kernel. Extremely lightweight. Used in Docker, Kubernetes. No separate OS per container.

6. Benefits:

- Server consolidation: Run many VMs on one physical server, reducing hardware costs.
- Isolation: Each VM is isolated; a crash in one does not affect others.
- Portability: VMs can be packaged and moved between hosts (OVF/OVA formats).
- Disaster Recovery: VM snapshots allow point-in-time recovery.
- Testing: Multiple OS environments on one machine for testing.

- Cloud Enablement: IaaS clouds are built on virtualised infrastructure.

Q26. Explain Cloud Service Models with Real Examples [12 Marks Marks]

Cloud service models define the level of abstraction and control provided to the customer. Real-world examples help illustrate the practical relevance of each model.

1. Infrastructure as a Service (IaaS) — In Depth:

IaaS provides virtualised computing infrastructure. The customer rents virtual machines, storage, and networking from a cloud provider. The provider handles physical data centres, hardware maintenance, and hypervisor management. The customer installs their own OS, middleware, and applications.

Real-world Example 1 – Netflix on AWS: Netflix uses Amazon EC2 (IaaS) to host its streaming servers. It provisions thousands of VMs during peak viewing hours and scales down during off-peak times, paying only for actual usage. This elasticity would be impossible to achieve with owned physical hardware.

Real-world Example 2 – Startup Web Hosting: A startup uses DigitalOcean Droplets (IaaS) to host its web application. Instead of buying physical servers, it rents virtual machines for a few dollars per month and scales as user traffic grows.

2. Platform as a Service (PaaS) — In Depth:

PaaS provides a managed environment for application development and deployment. The customer writes code and pushes it to the platform; the platform handles everything else—OS patches, scaling, load balancing, database management.

Real-world Example 1 – Google App Engine: A developer builds a Python Flask web app and deploys it to Google App Engine. GAE automatically handles load balancing, scaling (including to zero instances when no traffic), and monitoring. The developer only manages application code.

Real-world Example 2 – Heroku: A small team uses Heroku to deploy a Node.js REST API. They connect their GitHub repository, and Heroku automatically builds and deploys on every push. The team has no servers to manage.

3. Software as a Service (SaaS) — In Depth:

SaaS delivers complete software over the Internet. The user accesses it via a browser or app. There is no installation, no maintenance, and no version management required by the user. The provider manages everything.

Real-world Example 1 – Salesforce CRM: A sales team at a company uses Salesforce to manage customer relationships. They access it via a browser; there is no server to manage or software to install. Salesforce handles all uptime, security patches, and feature updates.

Real-world Example 2 – Microsoft 365: Employees use Word, Excel, Outlook, and Teams via Microsoft 365 (SaaS). Documents are stored in the cloud (OneDrive). Collaboration happens in real-time across locations with no software installation required.

4. Emerging Service Models:

- **FaaS (Function as a Service):** Serverless computing where code runs in response to events. Example: AWS Lambda. Developers write individual functions; the cloud provider handles execution, scaling, and billing per invocation.
- **IDaaS (Identity as a Service):** Cloud-based identity management. Example: Okta. Provides SSO and MFA without building identity infrastructure.

- **DBaaS (Database as a Service):** Managed database service. Example: Amazon RDS, MongoDB Atlas. No server management required for database operations.

5. Responsibility Matrix:

Layer	IaaS	PaaS	SaaS
Physical hardware	Provider	Provider	Provider
Virtualisation	Provider	Provider	Provider
OS	User	Provider	Provider
Runtime/Middleware	User	Provider	Provider
Application	User	User	Provider
Data	User	User	Provider

UNIT III: Cloud Platforms

Q27. Example of AWS / Azure / Google Cloud [2 Marks Marks]

- **AWS (Amazon Web Services):** Provides services like EC2 (compute), S3 (storage), RDS (database), Lambda (serverless). Example: Netflix uses AWS for video streaming.
 - **Microsoft Azure:** Offers Azure Virtual Machines, Azure Blob Storage, Azure SQL, Azure Active Directory. Example: LinkedIn runs on Azure.
 - **Google Cloud Platform (GCP):** Provides Compute Engine, Cloud Storage, BigQuery (analytics), GKE (Kubernetes). Example: Spotify uses GCP for data analytics.
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Q28. List Cloud Platforms [2 Marks Marks]

Major cloud platforms include:

- Amazon Web Services (AWS) – Market leader; broadest service portfolio.
 - Microsoft Azure – Strong enterprise and hybrid cloud offering.
 - Google Cloud Platform (GCP) – Leader in data analytics, ML, and Kubernetes.
 - IBM Cloud – Strong in AI (Watson) and enterprise applications.
 - Oracle Cloud – Focused on databases and enterprise workloads.
 - Alibaba Cloud – Dominant in Asia-Pacific region.
 - Salesforce / Force.com – PaaS and SaaS for CRM.
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Q29. Salesforce vs Force.com [4 Marks Marks]

Aspect	Salesforce	Force.com
Type	SaaS Application	PaaS Platform
Definition	Ready-to-use CRM application	Platform to build custom cloud apps
Target users	Sales, marketing, support teams	Developers and businesses
Customisation	Limited to configuration	Fully customisable via Apex code
Language	None (configuration)	Apex (Java-like), Visualforce, SOQL
Examples	Sales Cloud, Service Cloud, Marketing Cloud	Custom business apps on Salesforce infrastructure
Dependency	Runs on Force.com	Underlying platform of Salesforce

Q30. Describe Service-Oriented Architecture (SOA) in Cloud [4 Marks Marks]

Service-Oriented Architecture (SOA) is a design approach where software is built as a collection of interoperable services. In cloud computing, SOA provides the architectural foundation for building loosely coupled, reusable cloud services.

Key Principles of SOA:

- **Service Contract:** Each service has a well-defined interface (typically WSDL for SOAP or OpenAPI for REST).
- **Loose Coupling:** Services have minimal dependencies; changes to one service don't break others.
- **Abstraction:** Internal implementation details are hidden; only the interface is exposed.
- **Reusability:** Services are designed to be reused across different applications and contexts.
- **Statelessness:** Services are ideally stateless, storing no session information between calls.

SOA in the Cloud Context:

Cloud APIs (REST, SOAP, GraphQL) expose cloud services as SOA-compliant interfaces. Microservices architecture (a modern evolution of SOA) decomposes applications into small, independently deployable services, commonly hosted in containers (Docker/Kubernetes) on cloud platforms.

Q31. Explain AWS Architecture [7 Marks Marks]

Amazon Web Services (AWS) is the world's most comprehensive cloud platform, offering over 200 services. Its architecture is built around regions, availability zones, and a broad service portfolio.

Global Infrastructure:

- **Regions:** Geographic areas (e.g., US-East-1, AP-South-1 Mumbai) containing multiple data centres. As of 2024, AWS has 30+ regions globally.
- **Availability Zones (AZs):** Each region has 3–6 AZs—physically separate data centres within the region. Deploying across AZs ensures fault tolerance.
- **Edge Locations:** Globally distributed points of presence for CloudFront (CDN) to deliver content with low latency.

Core Service Categories:

- **Compute:** EC2 (VMs), Lambda (serverless), ECS/EKS (containers), Elastic Beanstalk (PaaS).
- **Storage:** S3 (object storage), EBS (block storage), EFS (file system), Glacier (archival).
- **Databases:** RDS (relational: MySQL, PostgreSQL, Oracle), DynamoDB (NoSQL), Redshift (data warehouse), ElastiCache (in-memory).
- **Networking:** VPC (Virtual Private Cloud), Route 53 (DNS), CloudFront (CDN), Direct Connect (dedicated links), ELB (load balancing).
- **Security and Identity:** IAM (identity and access management), KMS (key management), Shield (DDoS protection), WAF (web application firewall).
- **Analytics:** EMR (Hadoop/Spark), Athena (serverless SQL), Kinesis (real-time streaming), QuickSight (BI).

Well-Architected Framework:

AWS defines the Well-Architected Framework with six pillars: Operational Excellence, Security, Reliability, Performance Efficiency, Cost Optimization, and Sustainability. This framework guides organisations in building optimal cloud solutions on AWS.

Q32. Describe Google Cloud Platform (GCP) [7 Marks Marks]

Google Cloud Platform is Google's cloud computing suite, built on the same infrastructure Google uses for its own products like Search, Gmail, and YouTube.

Global Infrastructure:

GCP operates across 35+ regions and 100+ zones globally. Google's private fibre network connects its data centres, providing ultra-low latency between regions.

Key Services:

- **Compute Engine:** IaaS VM service. Supports custom machine types, preemptible instances, and sole-tenant nodes.
- **Google Kubernetes Engine (GKE):** Managed Kubernetes for containerised workloads. Google invented Kubernetes, so GKE is the industry benchmark.
- **App Engine:** Fully managed PaaS for web and mobile applications. Supports multiple languages (Python, Java, Go, PHP).
- **Cloud Storage:** Object storage (similar to S3) with four storage classes: Standard, Nearline, Coldline, Archive.
- **BigQuery:** Serverless, highly scalable data warehouse. Enables SQL queries on petabyte-scale datasets in seconds.
- **Cloud Spanner:** Globally distributed, strongly consistent relational database. Unique to GCP.
- **Vertex AI:** End-to-end AI/ML platform for training, deploying, and managing ML models.
- **Cloud Pub/Sub:** Asynchronous messaging service for event-driven architectures.

Strengths of GCP:

GCP is particularly strong in data analytics, machine learning, and Kubernetes-based container orchestration. Companies like Spotify, Twitter, and HSBC use GCP for large-scale data processing.

Q33. Explain Microsoft Azure Services [7 Marks Marks]

Microsoft Azure is the second-largest cloud provider, known for its strong enterprise integration, hybrid cloud capabilities, and Microsoft product ecosystem.

Global Presence:

Azure operates across 60+ regions and 140+ countries—more regions than any other cloud provider. Each region has multiple availability zones.

Core Services:

- **Azure Virtual Machines:** IaaS compute service supporting Windows and Linux VMs of various sizes.

- **Azure App Service:** PaaS for hosting web apps, REST APIs, and mobile backends with auto-scaling.
- **Azure Blob Storage:** Object storage for unstructured data. Tiers: Hot, Cool, Archive.
- **Azure SQL Database:** Fully managed relational database service based on Microsoft SQL Server.
- **Azure Cosmos DB:** Globally distributed, multi-model NoSQL database with five consistency levels.
- **Azure Active Directory (Entra ID):** Cloud-based IAM service for SSO, MFA, and conditional access. Widely used for enterprise identity.
- **Azure Kubernetes Service (AKS):** Managed Kubernetes for containerised application deployment.
- **Azure DevOps:** Full suite for CI/CD pipelines, source control (Git), testing, and project management.
- **Azure AI / Cognitive Services:** Pre-built AI APIs for vision, speech, language, and decision-making.

Hybrid Cloud Leadership:

Azure Arc extends Azure management to on-premises and multi-cloud environments. Azure Stack lets organisations run Azure services in their own data centres, making Azure the premier choice for hybrid cloud deployments.

Q34. Illustrate Aneka Cloud Integration [7 Marks Marks]

Aneka is a Platform as a Service (PaaS) framework developed by Manjrasoft that enables integration of private and public cloud resources into a unified computing environment. It is built on the .NET framework and supports multiple programming models.

Core Architecture:

- **Aneka Container:** The fundamental unit of deployment. Runs on each cloud node and provides application execution, resource management, and monitoring. Containers communicate with each other to form the Aneka cloud fabric.
- **Reservoir Manager:** Manages the pool of Aneka containers. Responsible for resource provisioning, scheduling, and scaling.
- **Application Repository:** Stores application binaries and configuration files. Enables deployment across distributed nodes.
- **Schedulers:** Multiple scheduling policies are supported (FIFO, SLA-based, priority-based). Users can plug in custom schedulers.

Programming Models Supported:

- **Thread Programming Model:** Maps .NET threads onto distributed cloud resources. Useful for parallel applications.
- **Task Programming Model:** Independent tasks are distributed across cloud nodes. Suitable for embarrassingly parallel workloads.
- **MapReduce Programming Model:** Aneka supports MapReduce for distributed data processing, similar to Hadoop.

Private + Public Cloud Integration:

Aneka's cloud bursting capability allows applications to overflow from a private cloud to a public cloud (e.g., Amazon EC2) when the private cloud is at capacity. This seamless integration is governed by SLA-aware scheduling, which decides whether to use private or public resources based on cost, performance, and availability constraints.

Advantages of Aneka:

- Language-agnostic (supports C#, .NET)
- Flexible and pluggable architecture
- Transparent cloud bursting
- Strong SLA management for enterprise workloads

UNIT IV: Cloud Migration & Security

Q35. Define Cloud Migration [2 Marks Marks]

Cloud migration is the process of moving an organisation's digital assets—applications, data, IT resources, and workloads—from on-premises infrastructure or from one cloud environment to another cloud environment. The goal is to leverage the scalability, flexibility, and cost benefits of cloud computing. Migration involves planning, assessment, data transfer, application refactoring (if needed), testing, and cutover.

Q36. Define Cloud Storage [2 Marks Marks]

Cloud storage is a model of data storage where digital data is stored on remote servers (data centres) managed by a cloud provider, and accessed over the Internet. It eliminates the need for local physical storage devices. Cloud storage is highly scalable, durable, and accessible from anywhere. Types include object storage (Amazon S3), block storage (Amazon EBS), and file storage (Amazon EFS, Google Filestore).

Q37. What is Cloud API? [2 Marks Marks]

A Cloud API (Application Programming Interface) is a set of protocols, routines, and tools that allow applications to interact programmatically with cloud services. Instead of using a GUI (web console), developers use APIs to automate provisioning, management, and scaling of cloud resources. Cloud APIs are typically RESTful (using HTTP methods: GET, POST, PUT, DELETE) and return responses in JSON or XML. Examples: AWS SDK, Google Cloud API, Azure REST API.

Q38. Explain Migration Strategies [4 Marks Marks]

The most widely adopted framework for cloud migration strategies is Gartner's '6 Rs' (later expanded to 7 Rs by AWS):

- **Rehost (Lift and Shift):** Move applications to the cloud without modification. Fastest migration approach. Best for legacy applications where re-architecture is not feasible. Example: Moving a VM from on-premises to an AWS EC2 instance.
 - **Replatform (Lift, Tinker, and Shift):** Make minor optimisations during migration without changing core architecture. Example: Moving a database to Amazon RDS (managed service) instead of running it on a VM.
 - **Refactor / Re-architect:** Redesign the application to be cloud-native (e.g., microservices, serverless). Highest effort but best long-term benefits in scalability and cost.
 - **Repurchase:** Replace existing software with a cloud-native SaaS alternative. Example: Replace on-premises CRM with Salesforce.
 - **Retire:** Identify applications that are no longer needed and decommission them.
 - **Retain:** Keep certain applications on-premises due to compliance, latency, or complexity constraints.
-

Q39. Explain Data Security Concepts [4 Marks Marks]

Data security in the cloud involves protecting data from unauthorised access, corruption, and loss throughout its lifecycle.

- **Encryption at Rest:** Data stored in cloud storage is encrypted using AES-256 or similar algorithms. Even if storage media is compromised, data is unreadable without the key. Example: AWS S3 SSE (Server-Side Encryption).
 - **Encryption in Transit:** Data is encrypted during transmission using TLS/SSL protocols. Prevents eavesdropping and man-in-the-middle attacks.
 - **Access Control (IAM):** Role-Based Access Control (RBAC) ensures users and services access only the resources they are authorised for. Principle of Least Privilege is applied.
 - **Data Masking / Tokenisation:** Sensitive data (e.g., credit card numbers, PII) is replaced with tokens or masked values in non-production environments.
 - **Data Integrity:** Checksums (MD5, SHA-256) and digital signatures verify that data has not been tampered with.
 - **Key Management:** Cryptographic keys are stored and rotated using dedicated Key Management Services (AWS KMS, Azure Key Vault).
-

Q40. Explain Cloud Migration Approaches (Detailed) [7 Marks Marks]

Cloud migration approaches describe the broad strategies organisations use when moving to the cloud. There are several major approaches, each suited to different organisational contexts.

1. Phased Migration:

Applications are migrated in stages, starting with less critical workloads to gain experience before migrating mission-critical systems. This reduces risk and allows the team to learn the cloud environment progressively.

2. Big Bang Migration:

All systems are migrated at once during a defined cutover window. This minimises the time spent running parallel environments but carries higher risk. Suitable for small organisations with simple IT landscapes.

3. Hybrid Migration:

Some workloads remain on-premises while others move to the cloud. The on-premises and cloud environments are integrated via a hybrid architecture (VPN, Direct Connect, Azure ExpressRoute). This approach accommodates compliance requirements or latency-sensitive applications.

4. Cloud-Native Rebuild:

Legacy applications are completely re-architected as cloud-native applications using microservices, containers, and serverless functions. This approach requires the most time and cost but delivers the greatest cloud benefits.

5. Data Migration Approaches:

- **Online Migration:** Data is transferred over the Internet or dedicated links. Suitable when downtime is acceptable.
- **Offline Migration:** Physical media (AWS Snowball) is used to transfer large volumes of data when network bandwidth is insufficient.

- **Replication:** Real-time data replication from source to cloud target, minimising downtime during cutover.

Q41. Describe Cloud Storage Systems [7 Marks Marks]

Cloud storage systems provide scalable, durable, and accessible storage over the Internet. There are three primary types of cloud storage:

1. Object Storage:

Stores data as objects with a unique identifier (key), metadata, and the data itself. There is no folder hierarchy—objects are accessed via HTTP APIs. Highly scalable and ideal for unstructured data.

- **Examples:** Amazon S3, Google Cloud Storage, Azure Blob Storage.
- **Use Cases:** Backup and archival, static website hosting, big data analytics, media storage.
- **Key Feature:** Virtually unlimited scale; objects can range from bytes to terabytes.

2. Block Storage:

Provides raw storage volumes that are formatted with a file system and attached to a VM as a virtual disk. Offers low-latency, high-performance access. Suitable for databases and applications requiring consistent I/O.

- **Examples:** Amazon EBS, Azure Managed Disks, Google Persistent Disk.
- **Use Cases:** Boot volumes for VMs, relational databases (MySQL, Oracle), transactional applications.

3. File Storage (NAS in the Cloud):

Provides shared file systems accessible by multiple VMs simultaneously using standard protocols (NFS, SMB/CIFS).

- **Examples:** Amazon EFS, Azure Files, Google Filestore.
- **Use Cases:** Shared content repositories, home directories, development environments.

4. Cloud Storage Features:

- **Durability:** Data is replicated across multiple devices and locations (e.g., Amazon S3 provides 11 nines of durability).
- **Availability:** Services are designed for 99.9%–99.99% availability SLAs.
- **Tiering:** Automatic storage class transitions (hot → cool → archive) to reduce costs.
- **Versioning:** Multiple versions of an object are maintained for recovery from accidental deletion.
- **Lifecycle Policies:** Automated rules to transition or delete data after defined periods.

Q42. Explain Role of APIs in Cloud [4 Marks Marks]

APIs (Application Programming Interfaces) are the primary mechanism through which cloud services are programmatically accessed and managed.

- **Resource Provisioning:** APIs allow developers to create, start, stop, and delete cloud resources (VMs, databases, storage buckets) via code rather than manual console interaction.

- **Automation and IaC:** Infrastructure as Code tools (Terraform, AWS CloudFormation, Pulumi) use cloud APIs to automate provisioning. This ensures reproducible, version-controlled infrastructure.
- **Integration:** APIs enable cloud services to integrate with on-premises systems, third-party services, and each other. Example: An application uses the S3 API to upload files and the SES API to send email notifications.
- **Monitoring and Management:** APIs provide access to metrics, logs, and events for automated monitoring and auto-scaling. Example: AWS Auto Scaling uses CloudWatch API to trigger scaling actions.
- **Security and Access Control:** IAM APIs enforce who can do what via programmatic policy management.

Q43. Explain Cloud Security Challenges & Solutions [7 Marks Marks]

Cloud security involves the policies, controls, and technologies that protect cloud computing environments, applications, and data.

Key Security Challenges:

- **Data Breaches:** Unauthorised access to sensitive data stored in the cloud. Caused by misconfigured storage buckets, weak credentials, or vulnerabilities.
- **Identity and Access Management (IAM) Misconfiguration:** Over-permissive roles and policies expose resources unnecessarily. The '2019 Capital One breach' was caused by an overly permissive IAM role.
- **Insecure APIs:** Poorly designed or unprotected APIs expose cloud services to attacks like injection, parameter tampering, and DoS.
- **Data Loss:** Accidental deletion, hardware failure, or ransomware attacks. Mitigated by versioning and cross-region replication.
- **Account Hijacking:** Stolen credentials allow attackers to control cloud accounts and resources.
- **Insider Threats:** Malicious or careless employees with cloud access can cause significant damage.
- **Compliance and Regulatory Risks:** Organisations must ensure their cloud deployments comply with GDPR, HIPAA, PCI-DSS, etc.

Solutions and Best Practices:

- **Encryption:** Encrypt all data at rest and in transit. Use customer-managed keys (CMK) for sensitive workloads.
- **Zero Trust Architecture:** Never trust, always verify. Authenticate and authorise every request regardless of source.
- **Principle of Least Privilege:** Grant users and services only the minimum permissions required.
- **Multi-Factor Authentication (MFA):** Require MFA for all user accounts, especially privileged ones.
- **Security Groups and Network ACLs:** Restrict network access at the VM and subnet level.
- **SIEM and Monitoring:** Use security information and event management tools to detect and respond to threats. Example: AWS CloudTrail + GuardDuty.
- **Regular Audits and Penetration Testing:** Proactively identify and remediate vulnerabilities.

Q44. Describe Seven-Step Cloud Migration Model [12 Marks Marks]

The Seven-Step Migration Model provides a structured, phase-by-phase approach to cloud migration. Each step builds on the previous one, ensuring a safe, well-planned transition.

Step 1: Assessment and Discovery

Conduct a comprehensive inventory of all existing IT assets—applications, servers, databases, network components, and dependencies. Classify workloads by complexity, business criticality, regulatory requirements, and migration readiness. Tools: AWS Migration Evaluator, Azure Migrate, CloudEndure.

Step 2: Business Case and Objectives

Define the business justification for cloud migration. Calculate total cost of ownership (TCO) for current infrastructure vs cloud. Identify key objectives: cost reduction, scalability, disaster recovery, innovation. Obtain stakeholder buy-in and executive sponsorship.

Step 3: Migration Strategy Selection

For each application, determine the appropriate migration strategy from the 6 Rs: Rehost, Replatform, Refactor, Repurchase, Retain, or Retire. Prioritise quick wins (Rehost) to build confidence and capability before tackling complex re-architecture.

Step 4: Proof of Concept (PoC)

Select a non-critical workload and migrate it to the cloud as a proof of concept. This validates the technical approach, tools, and processes before full-scale migration. Identify and resolve unexpected issues in a low-risk environment.

Step 5: Migration Planning

Create a detailed migration plan with timelines, resource allocation, dependencies, and rollback procedures. Set up the target cloud environment (networking, security, IAM, monitoring). Establish migration waves—groups of applications to be migrated together based on dependencies.

Step 6: Execution (Migration)

Execute the migration in waves. For each wave: migrate the application/data, run tests to validate functionality and performance, cut over traffic from the old environment to the new cloud environment. Use migration tools: AWS Database Migration Service (DMS), Azure Site Recovery, Google Transfer Appliance. Monitor closely during and after cutover.

Step 7: Optimisation and Continuous Improvement

After migration, focus on optimising the cloud environment. Right-size VM instances based on actual usage. Implement auto-scaling, reserved instances (for cost savings), and cloud-native services. Continuously monitor performance, costs, and security using cloud-native tools. Establish a Cloud Centre of Excellence (CCoE) for ongoing governance.

Summary Table:

Step	Phase	Key Activity
1	Assessment	Inventory all assets, classify workloads
2	Business Case	Define objectives, calculate TCO

Step	Phase	Key Activity
3	Strategy Selection	Choose from 6 Rs per application
4	Proof of Concept	Migrate a non-critical app first
5	Planning	Create migration waves, rollback plans
6	Execution	Migrate, test, cut over, monitor
7	Optimisation	Right-size, auto-scale, govern continuously

Q45. Explain Data Protection in Cloud [7 Marks Marks]

Data protection in the cloud encompasses all measures taken to ensure that data is accurate, available, and confidential throughout its lifecycle.

1. Data Classification:

Before protecting data, it must be classified by sensitivity level: Public, Internal, Confidential, Restricted. Different protection controls are applied based on classification.

2. Backup and Recovery:

Regular automated backups are essential. Cloud providers offer managed backup services (AWS Backup, Azure Backup). Recovery Time Objective (RTO) and Recovery Point Objective (RPO) define acceptable downtime and data loss, respectively. Backups should be tested regularly.

3. Disaster Recovery (DR):

DR strategies range from simple backups (cold DR) to active-active replication across multiple regions (hot DR). Cloud makes geo-redundant DR economically feasible for all organisations.

4. Compliance and Legal Protection:

GDPR mandates data residency requirements (EU citizens' data must stay in the EU). HIPAA requires healthcare data to be encrypted and access-logged. Cloud providers offer compliance certifications (SOC 2, ISO 27001, PCI-DSS) to demonstrate their security posture.

5. Data Lifecycle Management:

Policies govern how long data is retained and when it must be deleted. Automatic retention and deletion policies in object storage prevent unnecessary data accumulation and reduce compliance risk.

6. Immutable Storage:

WORM (Write Once Read Many) storage prevents data from being modified or deleted for a defined period. Used for audit logs and compliance records. Example: AWS S3 Object Lock, Azure Immutable Blob Storage.

UNIT V: Tools & Future Cloud

Q46. Define Eucalyptus [2 Marks Marks]

Eucalyptus (Elastic Utility Computing Architecture for Linking Your Programs To Useful Systems) is an open-source cloud computing platform for building private and hybrid clouds that are compatible with AWS. It implements AWS APIs, meaning applications developed for AWS EC2 and S3 can run on Eucalyptus without modification. Originally developed at UC Santa Barbara, it is one of the first open-source IaaS platforms.

Q47. Define Open Nebula [2 Marks Marks]

OpenNebula is an open-source cloud computing platform designed to build and manage private, public, and hybrid clouds. It provides a flexible solution for data centre virtualisation and cloud infrastructure management. OpenNebula supports multiple hypervisors (KVM, VMware, LXC) and integrates with public clouds (AWS, Azure) for hybrid deployments. It is known for its simplicity, flexibility, and adherence to open standards.

Q48. What is MapReduce? [2 Marks Marks]

MapReduce is a programming model and execution framework for processing and generating large datasets in parallel across a distributed cluster. It was introduced by Google and implemented in Apache Hadoop. The computation consists of two phases: the Map phase, where input data is split into key-value pairs and processed in parallel, and the Reduce phase, where intermediate results with the same key are aggregated to produce the final output. MapReduce is the foundation for big data processing in cloud environments.

Q49. Explain MapReduce Programming [7 Marks Marks]

MapReduce is a paradigm for distributed computation on large datasets. It abstracts the complexity of parallel computing, fault tolerance, and data distribution, allowing developers to focus on logic rather than infrastructure.

1. Core Concepts:

- **Input Splits:** The input data is divided into fixed-size chunks (e.g., 128 MB HDFS blocks). Each split is processed independently.
- **Map Function:** Processes each input split and emits intermediate key-value pairs. The logic (mapping) is user-defined. Example: For word count, the Map function emits (word, 1) for every word.
- **Shuffle and Sort:** The framework groups all values by key and sorts them. This happens automatically between Map and Reduce phases.
- **Reduce Function:** Aggregates all values for each unique key and produces the final output. Example: For word count, the Reduce function sums all values for each word.

2. Execution Flow:

- Step 1 – Input: Data is stored in HDFS (Hadoop Distributed File System) in blocks.
- Step 2 – Split: Input is divided into splits; each split assigned to a Map task.
- Step 3 – Map: Map tasks run in parallel on nodes where data resides (data locality).
- Step 4 – Shuffle/Sort: Intermediate output is partitioned, sorted, and transferred to Reducers.
- Step 5 – Reduce: Reduce tasks aggregate intermediate results.
- Step 6 – Output: Final output written to HDFS.

3. Word Count Example:

Input text: 'Cloud is great Cloud is scalable'

After Map: (cloud,1)(is,1)(great,1)(cloud,1)(is,1)(scalable,1)

After Shuffle: (cloud,[1,1])(great,[1])(is,[1,1])(scalable,[1])

After Reduce: (cloud,2)(great,1)(is,2)(scalable,1)

4. Fault Tolerance:

If a Map or Reduce task fails, the framework automatically re-schedules it on another node. This transparent fault handling is critical for processing on commodity hardware in large clusters.

5. Limitations and Evolution:

MapReduce is batch-oriented and not suitable for real-time processing. Modern frameworks like Apache Spark (in-memory), Apache Flink (stream processing), and Apache Storm have largely superseded Hadoop MapReduce for performance-critical workloads, but the MapReduce paradigm remains foundational.

Q50. Describe Dryad [4 Marks Marks]

Dryad is a distributed execution engine developed by Microsoft Research for running data-parallel programs on clusters. It provides a more flexible alternative to MapReduce by supporting general directed acyclic graphs (DAGs) of computation rather than limiting to just two phases (Map and Reduce).

Architecture:

- **Job Manager:** Central coordinator that constructs and manages the execution graph, monitors task progress, and handles failures.
- **Vertices:** Individual computation units (tasks) that perform data transformation. Vertices can execute any user-defined code.
- **Channels:** Data flows between vertices via channels—can be files, pipes, or shared memory, depending on whether vertices are on the same or different machines.

Advantages Over MapReduce:

- Supports arbitrary DAG topologies (not limited to two phases).
- Better suited for iterative algorithms (machine learning, graph processing).
- Efficient data pipelining via channels.

DryadLINQ:

DryadLINQ is a higher-level programming interface built on top of Dryad that allows developers to write distributed data processing programs using LINQ (Language Integrated Query) in C#. Queries are automatically compiled into Dryad execution graphs.

Q51. Define Open-Source Cloud Tools [2 Marks Marks]

Open-source cloud tools are publicly available, community-driven software platforms for building, managing, and deploying cloud infrastructure. Key examples include:

- **OpenStack:** Comprehensive IaaS platform for building private and public clouds.
 - **Eucalyptus:** AWS-compatible private cloud platform.
 - **OpenNebula:** Flexible cloud management platform supporting hybrid deployments.
 - **CloudStack:** IaaS platform used by large cloud providers (Exoscale).
 - **Kubernetes:** Open-source container orchestration platform by Google.
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Q52. Explain Eucalyptus, Open Nebula, and OpenStack (Detailed) [12 Marks Marks]

1. Eucalyptus

Eucalyptus (Elastic Utility Computing Architecture for Linking Your Programs To Useful Systems) is an open-source private cloud platform that implements AWS-compatible APIs, making it the first platform that allowed organisations to build private clouds that interoperate with AWS.

Architecture Components:

- **Cloud Controller (CLC):** Top-level management component. Handles authentication, resource pooling, and user management.
- **Walrus:** Object storage service compatible with Amazon S3. Stores VM images and data.
- **Cluster Controller (CC):** Manages network communication and VM scheduling within a cluster.
- **Storage Controller (SC):** Manages block storage, similar to Amazon EBS.
- **Node Controller (NC):** Runs on each physical host. Controls and monitors VM lifecycle on that node.

Key Features:

- AWS API compatibility (EC2, S3, IAM, Auto Scaling)
- Hybrid cloud capability with AWS
- Support for multiple hypervisors (KVM, VMware)
- Used for private cloud deployments in academic and research environments

2. OpenNebula

OpenNebula is a feature-rich, open-source cloud management platform designed for enterprise data centres. It supports multiple hypervisors, storage backends, and public cloud integrations.

Architecture:

- **OpenNebula Daemon (oned):** Core management service. Handles VM lifecycle, network management, storage, and user management.
- **Scheduler:** Determines which physical host should run each VM based on configured policies.
- **Drivers:** Pluggable components for virtualisation (KVM, VMware, LXC), storage (NFS, iSCSI, Ceph), and networking (VLAN, SDN).
- **Sunstone (Web UI):** Graphical interface for cloud administrators and users.

Key Features:

- Supports private, public, and hybrid cloud deployments
- Multi-cloud federation across multiple OpenNebula zones
- Integration with AWS, Azure, and GCP for hybrid cloud
- Strong focus on simplicity and ease of deployment
- Active community and regular releases

3. OpenStack

OpenStack is the most widely deployed open-source cloud infrastructure platform. It provides a comprehensive set of components to build IaaS clouds, used by major providers including Rackspace, OVHcloud, and Walmart.

Key Components:

- **Nova:** Compute service. Manages VM lifecycle across the cluster.
- **Swift:** Object storage, similar to Amazon S3.
- **Cinder:** Block storage, similar to Amazon EBS.
- **Neutron:** Networking service. Manages virtual networks, routers, and floating IPs.
- **Keystone:** Identity and authentication service. Provides IAM for all OpenStack components.
- **Glance:** Image service. Manages VM images (boot images for instances).
- **Horizon:** Web-based dashboard for managing OpenStack resources.
- **Heat:** Orchestration service using templates (similar to AWS CloudFormation).
- **Ceilometer:** Telemetry service for metering and billing.

Comparison Table:

Feature	Eucalyptus	OpenNebula	OpenStack
AWS Compatibility	Yes (EC2/S3)	No	Partial
Ease of Setup	Medium	Easy	Complex
Enterprise Use	Moderate	Yes	Very High
Hypervisors	KVM, VMware	KVM, VMware, LXC	KVM, Xen, VMware, LXC
Scalability	Medium	High	Very High
Community	Medium	Active	Very Large

Q53. Explain Future Trends in Cloud Computing [7 Marks Marks]

Cloud computing continues to evolve rapidly. Several key trends are shaping the future landscape:

1. Edge Computing:

Processing data closer to where it is generated (edge devices, IoT sensors) rather than in centralised cloud data centres. This reduces latency, bandwidth usage, and dependency on Internet connectivity. Key players: AWS Greengrass, Azure IoT Edge, Google Anthos.

2. Serverless Computing (FaaS):

Functions as a Service allows developers to write code that runs in response to events without managing any servers. The cloud provider handles all infrastructure, scaling, and availability. Cost is per invocation, not per instance. Example: AWS Lambda, Azure Functions, Google Cloud Run.

3. AI and ML as a Service:

Cloud providers offer managed AI/ML services that democratise machine learning. Organisations without deep ML expertise can use pre-trained models for vision, NLP, and prediction via APIs. Examples: AWS SageMaker, Google Vertex AI, Azure ML.

4. Kubernetes and Container-Native Cloud:

Kubernetes has become the standard for deploying and managing containerised applications at scale. Future clouds will be fully Kubernetes-native, with serverless containers (AWS Fargate, Cloud Run) abstracting even container management.

5. Multi-Cloud Strategy:

Organisations increasingly use services from multiple cloud providers to avoid vendor lock-in, optimise costs, and leverage best-in-class services. Multi-cloud management platforms (HashiCorp, Anthos, Azure Arc) are growing in importance.

6. Quantum Computing in the Cloud:

Cloud providers are making quantum computing hardware accessible as a cloud service. AWS Braket, Azure Quantum, and IBM Quantum allow researchers and enterprises to run quantum algorithms without owning quantum hardware.

7. Sustainable Cloud Computing:

Data centres consume significant energy. Cloud providers are investing in renewable energy, energy-efficient hardware, and carbon-neutral operations. Microsoft has pledged to be carbon-negative by 2030; Google has pledged 24/7 carbon-free energy by 2030.

Q54. Define the Mobile Market [2 Marks Marks]

The mobile market refers to the ecosystem of mobile devices (smartphones, tablets), mobile networks (4G/5G), mobile applications, and related services. It has grown explosively, with over 6 billion smartphone users globally. Mobile devices are the primary access point for Internet services in developing nations. The mobile market drives demand for cloud services because mobile apps require cloud-based backends for data storage, authentication, and computation. The mobile market is characterised by diverse device types, OS fragmentation (iOS vs Android), and varying connectivity conditions.

Q55. Using Smartphones with the Cloud [4 Marks Marks]

Smartphones are the primary front-end interface for cloud services for billions of users. The relationship between smartphones and the cloud is deeply integrated:

How Smartphones Use the Cloud:

- **Data Synchronisation:** Contacts, calendars, photos, and app data are automatically synchronised to the cloud (iCloud, Google Drive, OneDrive). Users seamlessly access their data on any device.
- **Cloud Storage:** Photos and documents are stored in the cloud rather than device storage, freeing local space. Example: Google Photos, Dropbox.
- **Cloud-Backed Applications:** Apps like Instagram, WhatsApp, Netflix, and banking apps process and store data in cloud backends. The smartphone is merely a thin client.
- **Mobile Backend as a Service (MBaaS):** Platforms like Firebase (Google) provide ready-made cloud backends for mobile apps—authentication, real-time database, push notifications.
- **Offloading Computation:** Computationally intensive tasks (speech recognition, image classification) are sent to cloud APIs rather than processed on the device, conserving battery and enabling more powerful AI.

Challenges:

Connectivity dependency, data privacy concerns, battery life for cloud sync, and varying network speeds (2G vs 5G) present challenges for consistent cloud-mobile experiences.
